Revisiting Methods to Assign Responses when Race and Hispanic Origin Reporting are Discrepant Across Administrative Records and Third Party Sources

by

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CES 24-26 **May 2024**

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Abstract

The Best Race and Ethnicity Administrative Records Composite file ("Best Race file") is a composite file which combines Census, federal, and Third Party Data (TPD) sources and applies business rules to assign race and ethnicity values to person records. The first version of the Best Race administrative records composite was first constructed in 2015 and subsequently updated each year to include more recent vintages, when available, of the data sources originally included in the composite file. Where updates were available for data sources, the most recent information for persons was retained, and the business rules were reapplied to assign a single race and single Hispanic origin value to each person record. The majority of person records on the Best Race file have consistent race and ethnicity information across data sources. Where there are discrepancies in responses across data sources, we apply a series of business rules to assign a single race and ethnicity to each record. To improve the quality of the Best Race administrative records composite, we have begun revising the business rules which were developed several years ago. This paper discusses the original business rules as well as the implemented changes and their impact on the composite file.

[*] Any opinions and conclusions expressed herein are those of the author and do not reflect the views of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data (Project No. 7509051, Disclosure Review Board (DRB) approval number: CBDRB-FY23-CES019-015).

Introduction

The Best Race and Ethnicity Administrative Records Composite file ("Best Race file") is an annual file which combines race and ethnicity responses from decennial census, household survey, federal administrative, and third-party data (TPD). The first version of the Best Race file was constructed using a combination of census, federal and third-party data sources (Ennis, et al. 2018). Since not all persons are captured by the decennial census or other Census Bureau survey data, and even those who do appear in Census data do not always have reported ethnicity and race information, combining Census data with other federal and third-party data sources creates a more comprehensive dataset with more complete information on individuals' race and ethnicity.

When administrative records are compiled, the majority of persons have consistent reports of race and ethnicity; however, race and Hispanic origin responses are not always the same for an individual across different administrative records (AR) sources. In the Best Race file, about three percent of individuals with a response for ethnicity have both Hispanic and non-Hispanic responses, and about three percent of individuals with a response for race have different race responses. To operationalize AR race and Hispanic origin data, we developed a set of business rules to assign a single "best" race and "best" Hispanic origin response when that information is discrepant across data sources.

The business rules originally developed for the Best Race file assigned Hispanic origin in the following way: for Hispanic origin, we first look for a code of "Hispanic" across all available data sources. If a person was coded as Hispanic in any data source, whether federal or TPD, then that record was assigned a value of Hispanic and would not be recoded to non-Hispanic regardless of any discrepancies across data sources. Where a Hispanic response was not present,

but a response of non-Hispanic could be found in a data source, then that person record was coded as non-Hispanic.

For race, we originally assigned race values in the following order: 1) Native Hawaiian or Pacific Islander alone; 2) American Indian or Alaska Native; 3) Two or More Races; 4) Asian alone; 5) Some Other Race alone; 6) Black alone; 7) White alone. Once a value is assigned to a record, it is not overwritten by other discrepant values.

The file was subsequently updated each year to include more recent vintages, when available, of the data sources originally included in the composite file. Where updates were available for data sources, the most recent information for persons was retained, and the business rules were reapplied to assign a single race and single Hispanic origin value to each person record.

Since its initial construction, the Best Race file has been used primarily in operations for the decennial census, but it has also been used for other research projects and to assist other programs. Among its applications, the Best Race file was used to assist in the production of a sample frame for the National Sample Survey of Registered Nurses. Records from the Best Race file were used to assign race and Hispanic origin information to persons included in the sample frame where that data was available in the Best Race file. Subsequent comparisons between the Hispanic origin values in the sample frame and responses obtained during the survey suggested inconsistencies between Hispanic origin in the frame and Hispanic origin as reported in the survey (Auerbach, Staiger, and Beurhaus 2021). Specifically, there was an over-representation in the sample frame of persons identified as Hispanic relative to the Hispanic population in the survey responses.

Based on these findings, we began to reassess the business rules as they relate to Hispanic origin and the use of third-party data. To improve the quality of the Best Race administrative records composite, we began revising the business rules which were developed for the first composite constructed in 2015. Here we will discuss changes to the business rules for the 2021 file build regarding the use of TPD files for the assignment of Hispanic origin as well as the construction of separate restricted and unrestricted composite files using Indian Health Services patient registration data.

Data and Methods

The Best Race composite file includes federal and third-party files to assign race and ethnicity to person records. Race is assigned using data from the decennial census, American Community Survey (ACS), the Social Security Administration Numerical Identification File (Numident), Medicare Enrollment Database, Housing and Urban Development (HUD) Tenant Rental Assistance Certification System, HUD Computerized Homes Underwriting Management System, HUD Public and Indian Housing Information Center, Indian Health Services, and Temporary Assistance for Needy Families. We use these same files plus the Medicaid Statistical Information System (MSIS) and four TPD sources to assign ethnicity as these additional files indicated higher levels of agreement with census data for Hispanic origin but not for race (Ennis et al. 2018). All records are processed through the Person Identification Validation System (PVS) which uses probability record linkage techniques to assign an anonymized Protected Identification Key (PIK) to each person record. This process allows for person records to be linked across datasets (Wagner and Layne 2014).

To ensure construction of the highest quality data product, we reevaluated the business rules used to assign Hispanic origin. In particular, we focused on the role of the Third-Party Data

files relative to other administrative records in the assignment of Hispanic origin. Two options were considered. The first option removes TPD completely from the Best Race file and would not consider TPD in assigning Hispanic origin just as the file does not use TPD to assign race. The second option considered would use TPD only as a last resort when no other data on Hispanic origin is available in any administrative records source file. The next section discusses the analysis of each option's treatment of Hispanic origin.

Results

2019 Best Race File Evaluation

To explore these options, the 2019 Best Race file was linked with the 2018 ACS and with the 2019 Current Population Survey (CPS).[1] While the ACS is used as an input for the Best Race file, the 2018 ACS was not used as an input for the 2019 Best Race file as it was not yet available. As such, there is not an issue of responses on the 2018 ACS informing Hispanic origin assignment in the 2019 Best Race file. The CPS is not an input file for the Best Race file.

Table 1 shows the response match rates for records which appear in both the Best Race file and also in the 2018 ACS and the 2019 CPS, and which have a valid value for Hispanic origin in each. ACS and CPS records were both limited to those with an unedited and unimputed Hispanic origin response. By limiting to unedited, unimputed responses, we can get a better comparison of responses without conflating any effects of the editing process in either survey. As shown in Table 1, when we either exclude TPD (Option 1) or use TPD as a last option for assigning Hispanic origin (Option 2), there are fewer records coded as Hispanic in the Best Race file but coded as non-Hispanic in the ACS (about 11 percent), relative to the original set of

[1] An analysis was also conducted using the 2017 ACS and 2018 CPS with similar results not shown here.

business rules (about 21 percent). For the CPS, the original set of business rules find about 16 percent of records assigned Hispanic in the Best Race data, but non-Hispanic in the CPS. Both Option 1 and Option 2 result in higher response match rates between the Best Race file and both the ACS and the CPS for persons who are assigned as being Hispanic compared to the response match rates for the current business rules. Option 1 yields the highest response match for persons coded as Hispanic, and Option 2 is slightly lower than Option 1. Agreement between the two improves in Options 1 and 2: about 9 percent of the time, the Best Race file codes a person as Hispanic when CPS reports them as non-Hispanic. The response match rates for persons coded as non-Hispanic in the Best Race file are consistently high across all comparisons.

Table 1. Best Race File linked to ACS and CPS: Comparison of Hispanic origin responses using Best Race as reference

		2018 ACS			2019 CPS		
2019 Best Race File		Hispanic	Non-Hispanic	Linked N	Hispanic	Non-Hispanic	Linked N
	Pct	Pct	Pct		Pct	Pct	
Current				4,069,000			135,000
Hispanic	100	78.9	21.1		83.8	16.2	
Non-Hispanic	100	1	99		1.5	98.5	
Option 1 - No TPD				4,005,000			133,000
Hispanic	100	89.1	10.9		91.1	8.9	
Non-Hispanic	100	1	99		1.6	98.4	
Option 2 - TPD Last				4,069,000			135,000
Hispanic	100	88.8	11.2		90.8	9.2	
Non-Hispanic	100	1	99		1.6	98.4	

Source: 2019 Best Race and Ethnicity File, 2018 ACS, and 2019 CPS

Table 2 displays match rates for records on the Best Race file that only have Hispanic origin data from the TPD. The response match rate is significantly lower for persons coded as Hispanic in TPD when compared to the ACS and CPS relative to the match rate for non-TPD

records. About one-third of persons assigned as being Hispanic through TPD have a corresponding response of non-Hispanic in the ACS and the CPS. However, two-thirds of records assigned as Hispanic through TPD have a matching survey response of Hispanic in the ACS or CPS. In practical terms, this is still better than flip-of-the-coin random selection. The TPD-ACS and TPD-CPS response match rate is still quite high for persons coded as non-Hispanic in TPD.

Table 2. Linked records on the 2019 Best Race File where Third Party Data is the only available data on Hispanic origin

		2018 ACS			2019 CPS		
2019 Best Race File		Hispanic	Non-Hispanic	Linked N	Hispanic	Non-Hispanic	Linked N
	Pct	Pct	Pct		Pct	Pct	
TPD Used to Assign				63,500			2,100
Hispanic	100	64	36		69.4	30.6	
Non-Hispanic	100	1.8	99		2.5	97.5	

Source: 2019 Best Race and Ethnicity File, 2018 ACS, and 2019 CPS

In terms of the coverage provided by the Best Race file, Option 1 (excluding TPD) would create an additional 6.5 million missing records on the Best Race file without a code assigned for Hispanic origin. Option 2, which only uses TPD as a last option when no other federal data sources have information on Hispanic origin, maintains the same number of records with a Hispanic origin code assigned as the current business rules. While Table 1 shows that there is a higher match rate for Hispanic origin using Option 1, that higher match rate comes at the expense of 6.5 million responses for Hispanic origin. Given that we know there is already lower coverage in administrative records for Hispanics relative to non-Hispanics (Rastogi and O'Hara 2012), we opt to prioritize coverage over the rate of response agreement between the composite

and survey records. Consequently, we chose to modify the business rules using Option 2, where we use TPD as a last resort when no other data on Hispanic origin is available in any administrative records source file. In order to give users the ability to prioritize agreement over coverage, we also created a flag on the file which will identify those records which rely on TPD for Hispanic origin. This will allow researchers the option to exclude records assigned Hispanic origin through TPD from analysis if they so choose.

To assess the effects of the changes in the business rules, we analyzed those records in the 2020 Best Race file that are impacted by the change in the treatment of TPD for assigning Hispanic origin. Table 3 shows the Hispanic-origin distribution among records which are assigned a value through the third-party data files as a last option. About 7.3 million records are assigned Hispanic origin through the TPD, and about 7 percent of these records are coded as Hispanic. There is no difference in the treatment of these records because of the change in business rules. In either set of business rules where the person record is present only in TPD and there is a discrepancy across multiple TPD sources, that record is assigned a value of Hispanic.

Table 3. 2020 Best Race File Hispanic Origin Distribution: Records Assigned by Third Party Data

	Revised Business Rules			
	Hispanic	Non-Hispanic	Total	
Original Business Rules	N	N	N	Percent
Hispanic	542,000	0	542,000	7.4
Non-Hispanic	0	6,772,000	6,772,000	92.6

Source: 2020 Best Race and Ethnicity Administrative Records Composite File, Restricted version

2020 Best Race File Evaluation

Table 4 presents the Hispanic origin distribution in the 2020 Best Race file according to the original business rules and the revised business rules which use TPD as a last option. The share assigned as Hispanic is lower under the revised business rules (17.9 percent) compared to the original business rules (19.3 percent).

Table 4. 2020 Best Race File Hispanic Origin Distribution: Original and Revised Business Rules

	Hispanic		Non-Hispanic		Total	
	N	Pct	N	Pct	N	Pct
Original Business Rules	67,170,000	19.3	281,700,000	80.7	348,900,000	100.0
Revised Business Rules	62,400,000	17.9	286,500,000	82.1	348,900,000	100.0

Source: 2020 Best Race and Ethnicity Administrative Records Composite File, Restricted version

We see in Table 5 that about 4.8 million records which would be assigned as Hispanic under the original business rules are instead assigned as non-Hispanic under the revised business rules. This change is a consequence of there being a response of non-Hispanic in one of the federal data sources and a code of Hispanic in the third-party data. With the TPD being used only as a last resort in the revised business rules, the person record is assigned as being non-Hispanic because of the federal data rather than Hispanic because of the third-party data.

Table 5. 2020 Best Race File Hispanic Origin Assignment: Original Compared to Revised Business Rules

	Revised Business Rules					
	Hispanic		Non-Hispanic		Total	
Original Business Rules	N	Pct	N	Pct	N	Pct
Hispanic	62,400,000	92.9	4,772,000	7.1	67,170,000	100.0
Non-Hispanic	0	0	281,700,000	100	281,700,000	100.0

To disaggregate those records which see a change in their Hispanic origin assignment because of the revised business rules, Table 6 shows the distribution of the Best Race assignment.

Among the records which switch from Hispanic to non-Hispanic under the revised rules, nearly 70 percent are White alone. The largest non-white group to switch from Hispanic to non-Hispanic is Asian alone (16.2 percent). Given that in the 2020 Census, only 0.4 percent of all Hispanics reported Asian as their race (U.S. Census Bureau, 2023), this suggests that the revised business rule likely results in a more accurate assignment of Hispanic origin for these records. Relative to all records in the Best Race file with a value for both race and Hispanic origin, there is a higher proportion of AIAN, NHPI, and Two or More Races among those who see a change from Hispanic to non-Hispanic under the revised rules.

Table 6. 2020 Best Race Assignment among Records Changing Hispanic Origin under Revised Business Rules

	Changed Hispanic Origin		All Best Race Records	
Best Race	N	Pct	N	Pct
White alone	3,297,000	69.4	288,100,000	74.3
Black alone	366,000	7.7	54,330,000	14.0
AIAN alone	86,500	1.8	4,209,000	1.1
Asian alone	769,000	16.2	14,450,000	3.7
NHPI alone	41,000	0.9	711,000	0.2
SOR alone	59,500	1.3	17,940,000	4.6
Two or More Races	130,000	2.7	7,981,000	2.1
Total	4,750,000	100.0	387,800,000	100.0

Source: 2020 Best Race and Ethnicity Administrative Records Composite File, Restricted version
Note: 22,500 records that changed Hispanic origin under the revised business rules do not have A race code assigned

Use of Indian Health Services Data

The Indian Health Services (IHS) Patient Registration file was included in the 2015-2019 versions of the Best Race and Ethnicity file. However, the data-use agreement for the 2019 and 2020 IHS files is more restrictive than for past data and limits the usage of the data. In order to make the Best Race file accessible to as many users as possible, we created two composites for the 2020 version – one version for groups with permission to access to the most recent IHS data delivery (i.e., "restricted"), and a second file which does not include the 2019 and 2020 IHS data but which still includes earlier IHS data (i.e., "unrestricted").

Table 7 shows the race distribution for both the restricted and unrestricted files. The restricted file has about 100,000 more records with race assigned than the unrestricted file, and conversely the unrestricted file has about 15,000 more records with missing race. Overall, the distribution is quite similar across the two files. As would be expected, the restricted file has a higher share of AIAN (1.09 percent) than does the unrestricted file (1.04 percent) since there are additional person records in the 2019 and 2020 IHS files which were not included in other deliveries. Over time, we can expect the gap in the share of AIAN between the two files to grow as new AIAN person records continue to be ingested into the restricted file but not into the unrestricted file.

In Table 8, we see the race distribution of records which are in both the restricted and unrestricted files and are also coded as AIAN in the restricted file. About 98 percent of these records are also coded as AIAN in the unrestricted file. However, about 1.8 percent have a different race code in the unrestricted file, and 0.4 percent are missing a race code in the unrestricted file. The implication here is that these PIKs were already on the AR composite file, however for the restricted file, they were also included in the 2019 or 2020 IHS delivery and subsequently were assigned as being AIAN.

Table 7. Best Race Distribution on Restricted and Unrestricted 2020 Updated Files

	Restricted		Unrestricted	
Best Race	N	Pct	N	Pct
White alone	288,100,000	74.3	288,200,000	74.3
Black alone	54,330,000	14.0	54,340,000	14.0
AIAN alone	4,209,000	1.1	4,040,000	1.0
Asian alone	14,450,000	3.7	14,450,000	3.7
NHPI alone	711,000	0.2	711,000	0.2
SOR alone	17,940,000	4.6	17,950,000	4.6
Two or More Races	7,981,000	2.1	7,989,000	2.1
Total	387,800,000	100.0	387,700,000	100.0
Missing	25,380,000		25,390,000	

Source: 2020 Best Race and Ethnicity Administrative Records Composite File, Restricted and Unrestricted versions

Table 8. Best Race Assignment Comparison of PIKs on Both Restricted and Unrestricted 2020 Updated Files

	Restricted File	
Unrestricted File	AIAN	
Best Race	N	Pct
White alone	56,000	1.4
Black alone	4,800	0.1
AIAN alone	4,040,000	97.9
Asian alone	1,900	0.0
NHPI alone	0	0.0
SOR alone	2,800	0.1
Two or More Races	7,800	0.2
Missing	14,500	0.4
Total	4,127,000	100

Source: 2020 Best Race and Ethnicity Administrative Records Composite File, Restricted and Unrestricted versions

Conclusion

The Best Race and Ethnicity Administrative Records Composite file continues to be a valuable resource to assist in Census Bureau operations and can contribute to household surveys and other research programs. Revisiting the business rules for the assignment of Hispanic origin and using Third Party Data files only when no other federal administrative records source has a response available will improve the quality of the data without losing any of its coverage. Including a flag on the file to indicate when TPD is used to assign Hispanic origin will give users the ability to decide whether or not to include those records depending on the scope of their research and any potential impact it may have for their work. Similarly, producing an unrestricted file which does not include 2019 and 2020 Indian Health Services data will allow the file to have a broader set of uses than the restricted file which is limited to decennial census research.

With the availability of the 2020 Census and other candidate administrative record sources, we will continue to revisit and revise the business rules for assigning race and Hispanic origin when responses are discrepant across different sources. We will also continue to evaluate additional administrative records sources with race and Hispanic origin data in the production of the composite file. By continuing to reassess the business rules and, where suitable, make changes, the Best Race file can continue to inform imputation strategies for nonresponse in the decennial census and in household surveys as well as assisting in the construction of survey frames and development of other research programs. Just as racial and Hispanic origin identification can be fluid, the business rules for assigning race and Hispanic origin in an AR composite must also be considered to be fluid to create the best possible measures and highest quality data.

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